Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/15/2025
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Analysis of ISS Data from the Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/01/2022	**End Date:**	06/30/2027
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2027 per F. Hernandez/NASA (Ed., 4/8/26). NOTE: End date changed to 06/30/2026 per J. McQuillen/GRC (Ed., 9/29/25). NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 10/24/24).		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Hasan, Mohammad Mujibul M.Sc. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC22K0328		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As space missions increase in scope, size, complexity, and duration, so do both power and heat dissipation demands. This is particularly the case for future manned missions to Mars. Paramount to the success of these missions is the ability to reduce size and weight, including those of thermal management sub-systems. One means to achieving this goal is to transition from single-phase to two-phase thermal management. By capitalizing upon the merits of both latent and sensible heat exchange rather than sensible exchange alone, two-phase systems can yield orders of magnitude enhancement in evaporation and condensation heat transfer coefficients compared to single-phase systems. These improvements are evident from recent NASA workshops that culminated in critical recommendations concerning the implementation of flow boiling and condensation in a variety of space applications such as Rankine cycle power conversion, thermal control systems, and advanced life support systems. The Flow Boiling and Condensation Experiment (FBCE) was conceived in 2011 with the intent of developing an integrated two-phase flow boiling/condensation facility for the International Space Station (ISS) to serve as a primary platform for obtaining two-phase flow and heat transfer data in microgravity. By comparing the microgravity data against those obtained in Earth's gravity, it will be possible to ascertain the influence of body force on two-phase transport phenomena in pursuit of predictive design tools, and to help determine minimum flow criteria that would ensure gravity independent flow boiling and condensation. FBCE is a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Key objectives of the proposed project will be to acquire information from the ISS microgravity heat transfer data and video records as well as assess validity and accuracy of recorded data for different operating conditions. The ISS data will be used to (1) assess and retrofit available empirical correlations and demonstrate validity for different gravities, (2) assess and retrofit available theoretical models and demonstrate validity for different gravities, (3) assess and retrofit models for minimum velocity that would ensure gravity independent heat transfer, and (4) develop computational fluid dynamics (CFD) models for both flow boiling and flow condensation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Advanced cooling methods for electric vehicles. 2. Cooling of fusion reactors. 3. Cooling of avionics. 4. Cooling of hybrid vehicle power electronics 5. Design of industrial heat exchangers.
Task Progress:	This study presents the flow condensation heat transfer results of the Flow Boiling and Condensation Experiment (FBCE). The primary goal of FBCE is to obtain fundamental flow boiling and condensation heat transfer data in microgravity (μge) through experiments onboard the International Space Station. Experiments were performed with the Condensation Module for Heat Transfer (CM-HT), which is a tube-in-tube counterflow heat exchanger. Condensing nPFH flows through a stainless steel tube with an inner diameter of 7.24 mm and rejects heat to cooling water flowing in an annular channel (with inner and outer gap diameter of 7.94 and 12.70 mm, respectively) surrounding the tube. Experiments tested a broad range of nPFH mass velocities, G = 72.8 – 291.5 kg/m2s, inlet thermodynamic equilibrium qualities, xe,in = 0.28 – 1.19, inlet pressures, pin = 103.9 – 160.2 kPa, and water mass velocities, Gw = 129.4 – 324.7 kg/m2s. A parametric investigation shows local condensation heat transfer coefficient, h, is primarily dependent on G and local xe, which can be represented by the two-phase mixture Reynolds number, Retp. Channel averaged heat transfer coefficient in the saturated two-phase region, $\bar{h}_{tp}$, increases with increasing G and xe,in. However, increasing inlet superheat does not affect $\bar{h}_{tp}$, but does increase the heat transfer coefficient averaged over the entire channel, $\bar{h}$. In the present experiments, G is sufficient to mitigate the effects of gravity, and $\bar{h}_{tp}$ in μge aligns with those for vertical down flow and horizontal flow in Earth gravity. Various correlations for htp were assessed, and the best performing correlation with a Mean Absolute Error (MAE) of 7.1% was that by Dorao and Fernandino, which is a function of Retp. Some correlations were shown to be overly dependent on the effect of gravity and were not applicable for the present μge database. A Separated Flow Model for annular condensation was employed to predict $\bar{h}_{tp}$. The model's physical basis makes it seamlessly adaptable for μge, and it resulted in a MAE of 32.3%.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Darges S, Mudawar I, Devahdhanush VS, Lee J, Kim S, Hasan MM, Nahra HK, Balasubramaniam R, Hall NR, Mackey J. "Flow Boiling and Condensation Experiment (FBCE): Latest findings from the summary ISS experiments." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Articles in Peer-reviewed Journals	Mudawar I, Darges SJ, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Experimental investigation and analysis of flow condensation heat transfer in microgravity–Experiments onboard the International Space Station." Int J Heat Mass Transf. 2026 Jan;254:127602. https://doi.org/10.1016/j.ijheatmasstransfer.2025.127602 , Jan-2026